UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 4: Systematic Review

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1. Mini Systematic Review

10 studies identified through documented search strings in Scopus and Semantic Scholar

Research question and scope

This mini systematic review examines how recent literature (2022–2026) proposes **reference architectures, enterprise architecture and architectural frameworks** to integrate **generative and agentic AI** (LLMs, agents, multi-agent systems) into **software development** in a governed way, and which **compliance, verification and governance** mechanisms they use. Guiding question:

What architectures and governance mechanisms have been proposed to ensure compliance when generative/agentic AI is used in the software development life cycle, and with what level of empirical validation?

The question is aligned with the doctoral problem of governing the use of AI in the development of solutions on top of an institutional reference architecture (application case: the Integrated Pension Information System (SIIP) of the Pension Normalization Office (ONP), whose Reference Architecture requires an Architectural Conformance Assessment (ECA) before development starts and explicitly includes artificial intelligence solutions in its scope).

Search strategy

Database: Scopus (TITLE-ABS-KEY fields). **Time filter:** PUBYEAR > 2021. **Result:** 21 documents. **Execution date:** September 2026.

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TITLE-ABS-KEY(("generative AI" OR genAI OR "large language model*" OR LLM* OR "agentic AI" OR "AI agent*" OR "LLM agent*" OR "multi-agent") AND ("reference architecture*" OR "enterprise architecture" OR "architectural framework*" OR TOGAF OR ArchiMate) AND ("software development" OR "software engineering" OR "code generation" OR SDLC) AND (complian* OR regulat* OR verif* OR validat* OR governance)) AND PUBYEAR > 2021
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The string combines four conceptual blocks (generative/agentic AI · reference/enterprise architecture · software development · compliance/governance) joined with AND. The synonyms inside each block are joined with OR, and wildcards are used to cover word variants.

Database: Semantic Scholar. **Time filter:** publication year 2022–2026 (equivalent to PUBYEAR > 2021 in Scopus). **Result:** 2 documents. **Execution date:** October 2026.

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("generative AI" | genAI | "large language model" | "large language models" | LLM* | "agentic AI" | "AI agent" | "AI agents" | "LLM agent" | "LLM agents" | "multi-agent") + ("reference architecture" | "reference architectures" | "enterprise architecture" | "architectural framework" | "architectural frameworks" | TOGAF | ArchiMate) + ("software development" | "software engineering" | "code generation" | SDLC) + (complian* | regulat* | verif* | validat* | governance)
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Peruvian databases

ALICIA — CONCYTEC

Filter: (arquitectura de referencia and TOGAF and inteligencia artificial generativa)

Result: zero (0)

RENATI — SUNEDU

Filter: (arquitectura de referencia and TOGAF)

Result: zero (0)

Eligibility criteria and selection procedure

Inclusion	Exclusion
IC1. It addresses generative AI, LLMs or AI agents. IC2. It proposes, evaluates or discusses a reference architecture, enterprise architecture, framework or architectural patterns. IC3. It is set in software engineering / SDLC or in the enterprise integration of these systems. IC4. It deals with compliance, verification, security or governance. IC5. Published between 2022 and 2026 and peer reviewed (article or conference paper).	EC1. Duplicates. EC2. Records without accessible text or without abstract. EC3. Works that use AI only as an analysis tool, with no architectural or governance component. EC4. Language other than English or Spanish. EC5. Grey literature, theses or chapters without peer review.

Procedure.

- Scopus returned 21 documents.
- Semantic Scholar returned 2 documents.
- ALICIA returned 0 (zero) documents.
- RENATI — SUNEDU returned 0 (zero) documents.

The 23 records were screened against IC1–IC5 and EC1–EC5 using title and metadata. The result was 10 included studies (meeting the deliverable's requirement of analyzing 10 papers), 4 records on hold because IC3/IC4 are not evident (abstract pending verification) and 9 excluded. Record-level detail is in Table 1 and the totals are in the PRISMA diagram.

Search results (n = 23)

Table 1 consolidates the 23 records identified: 21 from Scopus (in the default listing order) and 2 from Semantic Scholar. ALICIA–CONCYTEC and RENATI–SUNEDU returned 0 records. Each record was screened against criteria IC1–IC5 and EC1–EC5 using title and metadata, because the export does not include abstracts. Green: selected; yellow: meets what can be verified but is kept on hold; no color: excluded.

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
1	Jeong, Cho & Woo (2027). Architecting Reliable LLM-Integrated Simulation: From Code Generation to Tool-Augmented Interaction in Production Planning. IFIP Advances in Information and Communication Technology, 811.	Scopus	Conference paper	0	Excluded: IC5 (year 2027 in Scopus, outside 2022–2026); IC4 not evident.
2	[No authors] (2027). 25th International Conference on Perspectives in Business Informatics Research, BIR 2026. Lecture Notes in Business Information Processing, 601.	Scopus	Conference proceedings	0	Excluded: EC2 (proceedings without authors or study abstract).

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
3	Do Val Lopes, Pitta, Belem, Alves & Martins (2026). Engineering AI Agents for Clinical Workflows: A Case Study in Architecture, MLOps, and Governance. Proc. IEEE/ACM 5th Int. Conf. on AI Engineering (CAIN 2026).	Scopus	Conference paper	0	Selected (Table 2, #1).
4	Bandeira, De Oliveira, Maia, Aderaldo & Mendonça (2026). LLM-Native Applications: Toward a New Paradigm in Software Engineering. ICSE Companion 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
5	Alenezi, Akour & Al-Omari (2026). Agentic Artificial Intelligence in Software Engineering: Integration Points, Architectural Frameworks, Empirical Evaluation, and Governance. International Journal of Technology, 17(4).	Scopus	Article	1	Selected (Table 2, #2).
6	Elsisi, Santos & Melo (2026). Lifecycle-Aware GenAI Assistance with MCP via Context Refinement Loops: A Reference Architecture. FSE Companion 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
7	Stein (2026). Toward a taxonomy of generative AI use cases in business contexts: Integrating complexity, risk, and strategy. Journal of Computational Science, 96, 102826.	Scopus	Article	4	Selected (Table 2, #3).
8	Kushnerov, Sokol, Akhiiezer, … Nadtochyi & Balagura (2026). Zero-trust Conceptual Architecture for Secure Integration of Large Language Models and Autonomous AI Agents. ICHORA 2026.	Scopus	Conference paper	2	Selected (Table 2, #4).
9	Aldas-Alarcón & Carrera (2026). Mitigation of Artificial Intelligence Technical Debt and Its Impact on the Success of Institutional Information Systems During the Digital Transformation of Higher Education. ICEDEG (Part F228401).	Scopus	Conference paper	0	Excluded: IC1/IC2 (AI in general, not generative AI/LLM/agents; no explicit architectural component).
10	Siddhu, Pappu & Thanvi (2026). Compliance-Aware Generative AI for Life Insurance Document Synthesis: A Reference Architecture. ICAISET 2026.	Scopus	Conference paper	—	Selected (Table 2, #5).
11	Karuppuchamy (2026). Governed Agentic AI for Software Platforms: A Reference Architecture for Safe Autonomy at Scale. IEEE SoutheastCon 2026.	Scopus	Conference paper	1	Selected (Table 2, #6).
12	Moriarty (2026). From Classroom to Enterprise: Architectural Patterns for LLM Integration. IEEE SysCon 2026.	Scopus	Conference paper	0	Not selected: IC4 not evident in title/metadata (on hold; verify abstract).
13	Enyedi (2026). Human-Certified Module Repositories for the AI Age. IEEE AQTR 2026.	Scopus	Conference paper	1	Selected (Table 2, #7).
14	Pokala (2026). Agentic Dashboard Generation from Natural Language on Lakehouse Platforms. ICAISET 2026.	Scopus	Conference paper	0	Excluded: EC3/IC4 (AI used as a dashboard-generation tool; no governance component).
15	Moraes, Onofre & Aguiar (2026). Safeguarding Big Data Question-Answering Systems with a Two-Layer Input Guardrails. ICEIS 2026, vol. 1.	Scopus	Conference paper	0	Selected (Table 2, #8).
16	Jikeli (2026). Model-Driven Development and Low-Code Platforms in Healthcare Digitalization: Opportunities and Challenges. In Advancements in Digital Health and Care.	Scopus	Book chapter	1	Excluded: EC5 (book chapter); IC1 (does not address generative AI/LLMs/agents).
17	Zeydan & Aydeger (2026). Bias-Free and Auto-Evolving Generative AI: Design Principles, Architectures, and Reinforcement Integration. IEEE CCNC 2026.	Scopus	Conference paper	1	Selected (Table 2, #9).
18	Gadani (2026). Next-Generation Cloud Automation with Generative AI. In Revolutionizing the Cloud: Generative AI, Security and Sustainability.	Scopus	Book chapter	1	Excluded: EC5 (book chapter).

#	Reference (authors, year, title, source)	Database	Type	Cit.	Screening decision
19	Aminiranjbar, Tang, Wang, Pant & Viswanathan (2025). DAWN: Designing Distributed Agents in a Worldwide Network. IEEE Access, 13.	Scopus	Article	7	Not selected: IC3/IC4 not evident in title/metadata (on hold; verify abstract).
20	Joshi (2024). Secure Software Architecture for Enterprise Generative Artificial Intelligence. Int. Scientific and Technical Conf. on Computer Sciences and Information Technologies.	Scopus	Conference paper	2	Selected (Table 2, #10).
21	[No authors] (2024). 14th International Conference on Software Business, ICSOB 2023. Lecture Notes in Business Information Processing, 500.	Scopus	Conference proceedings	—	Excluded: EC2 (proceedings without authors or study abstract); IC1.
22	Vasa (2024). Generative AI and Agentic Orchestration for Data Engineering in Multi-Domain Enterprise Analytics Platforms. International Journal of Artificial Intelligence, Data Science, and Machine Learning, 5(4).	Semantic Scholar	Article	—	Excluded: IC4 not evident (data-quality validation, not AI governance); peer review to be confirmed (EC5).
23	Maire, Badmaev, … & Tirchas (2026). Agentic AI Framework for Technical Excellence: A Scalable Discipline-Based Multimodal Assistant for the Subsurface. IPTC Summit on AI for the Energy Industry.	Semantic Scholar	Conference paper	—	Excluded: IC3/IC4 (subsurface/energy domain; no software engineering or governance focus).

Table 1. Search results and screening decision. Cit. = Scopus citations (— = no data). Source: own elaboration based on Scopus and Semantic Scholar.

Selected studies (n = 10)

Table 2 presents the 10 studies that meet IC1–IC5 and do not fall under EC1–EC5. Records #4, #6, #12 and #19 of Table 1 meet the remaining criteria, but IC4 (or IC3) is not evident from title and metadata; they are kept on hold until verified against the abstract.

#	Reference	II1	II2	II3	II4	II5	Contribution / focus (basis of reading)
1	Do Val Lopes, Pitta, Belem, Alves & Martins (2026). Engineering AI Agents for Clinical Workflows: A Case Study in Architecture, MLOps, and Governance. CAIN 2026. Conference paper.	✓	✓	✓	✓	✓	AI agents in clinical workflows: case study on architecture, MLOps and governance (by title). Regulated domain (healthcare).
2	Alenezi, Akour & Al-Omari (2026). Agentic Artificial Intelligence in Software Engineering: Integration Points, Architectural Frameworks, Empirical Evaluation, and Governance. International Journal of Technology, 17(4). Article.	✓	✓	✓	✓	✓	Framework to integrate agentic AI into development; systematic review and multi-case analysis of 4 enterprise projects; reports no compliance breaches.
3	Stein (2026). Toward a taxonomy of generative AI use cases in business contexts: Integrating complexity, risk, and strategy. Journal of Computational Science, 96, 102826. Article.	✓	✓	✓	✓	✓	Four-layer taxonomy of complexity (A–D) and five perspectives, including technical autonomy and data governance.
4	Kushnerov, Sokol, Akhiiezer, … Nadtochyi & Balagura (2026). Zero-trust Conceptual Architecture for Secure Integration of Large Language Models and Autonomous AI Agents. ICHORA 2026. Conference paper.	✓	✓	✓	✓	✓	Zero-trust conceptual architecture to securely integrate LLMs and autonomous agents (by title).
5	Siddhu, Pappu & Thanvi (2026). Compliance-Aware Generative AI for Life Insurance Document Synthesis: A Reference Architecture. ICAISET 2026. Conference paper.	✓	✓	✓	✓	✓	GenAI reference architecture that is aware of compliance in a regulated domain (life insurance) (by title).
6	Karuppuchamy (2026). Governed Agentic AI for Software Platforms: A Reference Architecture for Safe Autonomy at Scale. IEEE SoutheastCon 2026. Conference paper.	✓	✓	✓	✓	✓	Reference architecture for governed agentic AI on software platforms (by title).

#	Reference	II1	II2	II3	II4	II5	Contribution / focus (basis of reading)
7	Enyedi (2026). Human-Certified Module Repositories for the AI Age. IEEE AQTR 2026. Conference paper.	✓	✓	✓	✓	✓	Human-certified module repositories, with provenance and interface contracts; reference design, vulnerability analysis and governance (abstract on arXiv:2603.02512).
8	Moraes, Onofre & Aguiar (2026). Safeguarding Big Data Question-Answering Systems with a Two-Layer Input Guardrails. ICEIS 2026, vol. 1. Conference paper.	✓	✓	✓	✓	✓	Two-layer input guardrail architecture to protect question-answering systems over big data (by title).
9	Zeydan & Aydeger (2026). Bias-Free and Auto-Evolving Generative AI: Design Principles, Architectures, and Reinforcement Integration. IEEE CCNC 2026. Conference paper.	✓	✓	✓	✓	✓	Design principles and architectures for bias-free, self-evolving generative AI (by title).
10	Joshi (2024). Secure Software Architecture for Enterprise Generative Artificial Intelligence. Int. Scientific and Technical Conf. on Computer Sciences and Information Technologies. Conference paper.	✓	✓	✓	✓	✓	Secure software architecture for enterprise generative AI (by title).

Table 2. Studies selected according to the inclusion criteria. IC1: generative AI/LLM/agents; IC2: architecture, framework or patterns; IC3: software engineering or enterprise integration; IC4: compliance, verification, security or governance; IC5: 2022–2026, peer reviewed. Source: own elaboration.

Synthesis of findings

• **Architectural proposals predominate.** 8 of 10 studies present an architectural artifact (reference or conceptual architecture, framework or reference design): #2, #4, #5, #6, #7, #8, #9 and #10; in 6 of them (#4, #5, #6, #8, #9 and #10) this is inferred from the title only. #1 is a case study on architecture and governance, and #3 is a taxonomy of use cases, with no architectural artifact.

• **Governance and compliance are explicit in 6 of 10 studies.** They appear in #1, #2, #3, #5, #6 and #7: agent governance (#1, #6), a framework with compliance evaluation (#2), risk and data governance (#3), regulatory compliance (#5) and certification with provenance (#7). The other four address security or mitigation of technical risks: zero-trust (#4), input guardrails (#8), secure software architecture (#10) and bias (#9).

• **Empirical evidence is scarce or cannot be verified.** Only #2 reports an identifiable empirical evaluation (multi-case analysis of 4 enterprise projects, with no compliance breaches and with stated limits to generalization). #7 presents a vulnerability analysis with no declared empirical evaluation, and for #4, #5, #6, #8, #9 and #10 validation cannot be verified because only the title was read.

• **Application contexts.** Enterprise and commercial settings predominate; the explicit regulated domains in the titles are healthcare (#1) and insurance (#5). Only #2 expressly addresses the use of agentic AI within software engineering; in #1, #4, #5, #8, #9 and #10 the AI is the system to be governed or protected, not support for development (by title). No title refers to digital government, the public sector or social security.

• **Maturity of the field.** 9 of the 10 studies are from 2026 and one from 2024 (#10); there are 8 conference papers and 2 journal articles (#2 and #3); Scopus citations range from 0 to 4 (no data for #5). It is an emerging field that is still not consolidated.

Limitations of this review

(i) Database coverage: only Scopus and Semantic Scholar; ALICIA–CONCYTEC and RENATI–SUNEDU returned 0 records. (ii) Screening of the 23 records was done on title and metadata only (the export does not include abstracts) and by a single reviewer; the 4 records on hold (Table 1, #4, #6, #12 and #19), including Elsisi et al. (2026) and Bandeira et al. (2026), which are close to the software life cycle, could change the synthesis. (iii) For 7 of the 10 included studies, reading was limited to title and metadata, so their contribution and empirical validation are inferred and the synthesis is provisional. (iv) Grey literature, theses and chapters without peer

review were excluded (EC5) and language was limited to English and Spanish (EC4), which leaves out AI governance standards and guidelines (e.g., ISO/IEC 42001, NIST AI RMF). (v) The search was run between September and October 2026.

Access links to the studies

- #1 Do Val Lopes et al. (2026) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105052340880
- #2 Alenezi et al. (2026) – Direct access: https://ijtech.eng.ui.ac.id/article/view/8380; Scopus: https://www.scopus.com/pages/publications/105046214271
- #3 Stein (2026) – Direct access: https://www.sciencedirect.com/science/article/pii/S187775032600044X; Scopus: https://www.scopus.com/pages/publications/105034631140
- #4 Kushnerov et al. (2026) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105042030100
- #5 Siddhu et al. (2026) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105042158521
- #6 Karuppuchamy (2026) – Direct access: https://ieeexplore.ieee.org/document/11475963; Scopus: https://www.scopus.com/pages/publications/105037800300
- #7 Enyedi (2026) – Direct access: https://arxiv.org/abs/2603.02512; Scopus: https://www.scopus.com/pages/publications/105045435826
- #8 Moraes et al. (2026) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105044791237
- #9 Zeydan & Aydeger (2026) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105034149112
- #10 Joshi (2024) – Direct access: not verified; Scopus: https://www.scopus.com/pages/publications/105005829833
- On hold (Table 1, #4) Bandeira et al. (2026) – Direct access: https://doi.org/10.1145/3774748.3795680; Scopus: https://www.scopus.com/pages/publications/105051355847
- On hold (Table 1, #6) Elsisi et al. (2026) – Direct access: https://doi.org/10.1145/3803437.3805617; Scopus: https://www.scopus.com/pages/publications/105045840085

Note. Numbering follows Table 2; the on-hold records cited in the text are identified by their number in Table 1. Scopus links come from the search export and require institutional access. “Not verified” means the publisher page or DOI has not yet been confirmed. The IEEE Xplore links recorded earlier (/document/11537001, /document/11541931, /document/11503623 and /document/11577868) are not assigned to any study until they are verified.

2. PRISMA 2020 Flow Diagram

Diagram built with the official PRISMA 2020 template for new systematic reviews which included searches of databases and registers only (Page et al., 2021), translated into English from the Spanish version. Its phases (identification; screening, which includes reports sought and assessed for eligibility; and inclusion) have non-empty values that are consistent with one another.

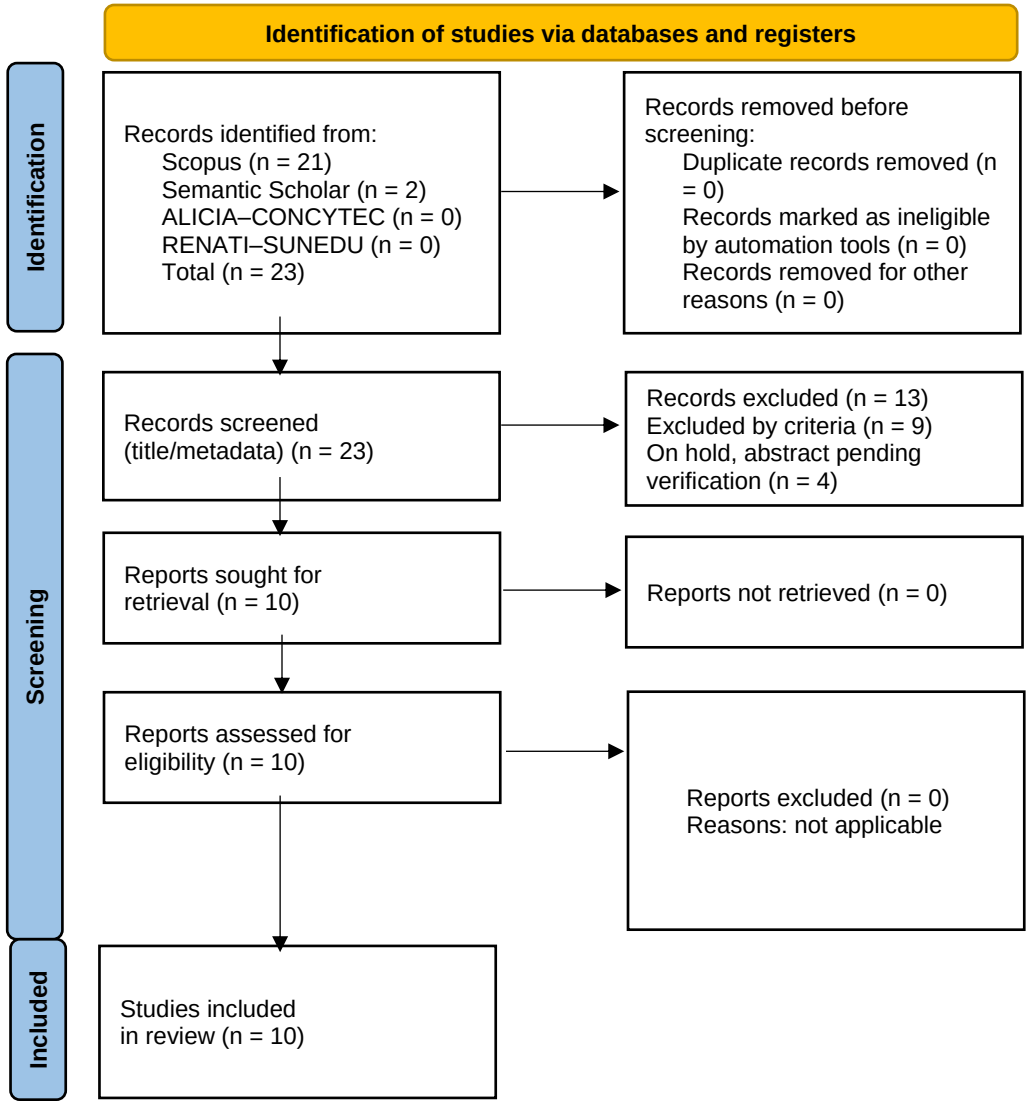


Figure 1. PRISMA 2020 flow diagram of the mini systematic review. Source: own elaboration, adapted from Page et al. (2021).

3. Gap Analysis Table

At least one candidate gap per type (knowledge · methodological · contextual · theoretical), with a citation as evidence.

At least one candidate gap was identified for each type.

Gap type	Candidate gap	Evidence (citation)	Research opportunity
Knowledge	There is no evidence on whether the use of agentic AI in the SDLC remains compliant with formal standards and architectures in contexts other than private enterprise projects.	Alenezi et al. (2026): report that "no compliance breaches were observed" in 4 enterprise projects, but warn that the findings are "context-specific and derived from a limited number of enterprise projects".	Study AI compliance in development under a mandatory architectural framework, in a public entity with multiple software factories.
Methodological	Architecture proposals with weak or undeclared empirical validation predominate; the few evaluations use few cases and quasi-experimental designs.	Alenezi et al. (2026): results "indicative rather than broadly generalizable", with a need for stronger quasi-experimental validation. Enyedi (2026): the abstract presents a reference design and vulnerability analysis, with no declared empirical evaluation. Elsisi et al. (2026), Siddhu et al. (2026) and Karuppuchamy (2026) are titled "reference architecture" (to be confirmed).	Design Science Research approach: build a governance artifact and evaluate it in cycles, with compliance metrics (e.g., conformance score) and before/after or control-group comparison.
Contextual	The corpus is concentrated on enterprise and commercial settings; the public sector and social security systems, with legacy system coexistence and State interoperability obligations, do not appear.	Alenezi et al. (2026): enterprise projects. Siddhu et al. (2026): life insurance, and Do Val Lopes et al. (2026): healthcare, the only explicit regulated domains. Stein (2026): business contexts. None of the 10 titles mentions digital government, the public sector or social security.	SIIP–ONP case: AI in development under the Reference Architecture, with the ECA, legacy coexistence and Peruvian digital government regulations. Verify the gap with the 13 records not included (4 on hold and 9 excluded) and with additional databases.
Theoretical	There are autonomy/risk taxonomies and governance proposals, but no theory linking AI autonomy levels to conformance criteria of a reference architecture (TOGAF/ArchiMate).	Stein (2026): A–D taxonomy and the distinction of layer D as "a fundamental phase change in system autonomy"; its abstract does not relate it to enterprise architecture frameworks. Karuppuchamy (2026) and Kushnerov et al. (2026) propose governed/zero-trust architectures (by title), with no declared mapping to architectural conformance.	Theoretical framework that associates each autonomy level (A–D) with controls, artifacts and architectural governance milestones (e.g., the ECA dimensions D1–D5), and subjects it to validation.

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